ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\ \text{m})/(5\times10^{-6}\ \text{m}) = 3200$. The metres cancel so the answer is in pixels. $\mathbf{f_x = 3200\ px}$.

(ii) Width of the sensor = N·p = 1280 × 5 μm = 6400 μm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\ \text{mm})(400\ \text{mm})/(16\ \text{mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 μm** on the belt.

(b) LSB = 5/4096 = 1.2207 mV.

ΔF = LSB / S = 1.2207 mV / 10 mV/N = **0.0122 N**.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}\,x(kT) + \int_0^T e^{A(T-s)}\,B\,u(kT)\,ds$. Sence u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}\,B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$$\mathbf{x_{k+1} = 0.8187\ x_k + 0.0906\ u_k}$$

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally because the derivitive of a sampled signal is not defined.

(b) ω_c = 20 rad/s. Tustin with T = 0.001: $a = \omega_c T/2 = 0.01$.

$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 - a)/(1 + a)]\,y_{k-1} = \mathbf{0.00990\ (u_k + u_{k-1}) + 0.98020\ y_{k-1}}$.
DC gain: 2(0.0099)/(1 − 0.9802) = 0.0198/0.0198 = 1 ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$.
Desiered $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$\mathbf{L = [18, -64]^T}$.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurment y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c)$ = 8 ms for its noise rejection, a trade-off it cannot escape. In addition the obsever gives all states, and can be augmented with a disturbance state

(e.g. unknown load torqe) which no filter on θ can estimate.

Circumstance where the fliter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The obsever propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free fliter just smooths what the encoder actually measures.

Sampling at 1 kHz is fast compared with the 10 rad/s observer bandwidth, so discretisation effects are small.